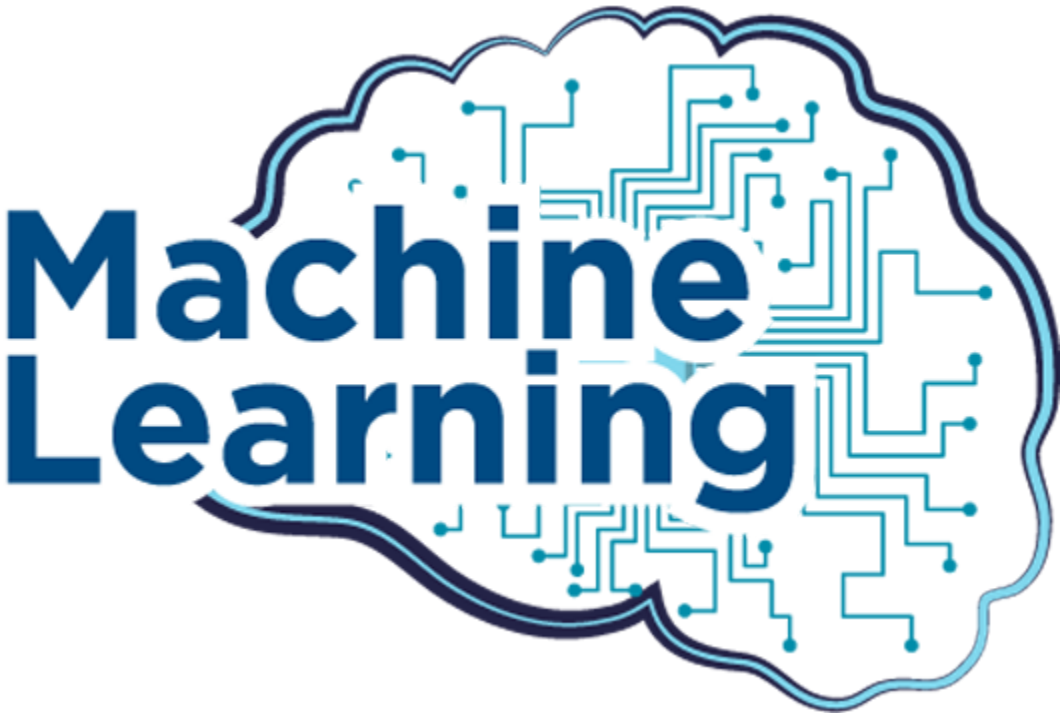


Proposal : IoT Machine Learning Project

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1. Project Idea

IoT Sentinel is an intrusion detection prototype implemented in software that can be used to detect possible attacks on networks in the Internet of Things environment. Smart cameras, sensors, and appliances for the home connected via a network may fall victim to cyber attacks.

The system will utilize machine learning algorithms to classify the network traffic as either Normal or Attack. The project will be developed in Python language using the free IoT Cybersecurity dataset. At the final stage, the Python program will be used to replay some data from the IoT dataset to create a stream of network traffic. A dashboard will be used to display the records and the predicted results by the model.
The target users of the project are students and developers who would like to learn more about using machine learning for basic IoT network monitoring.

2. Problem Statement

IoT networks can be targeted by different cyberattacks, including denial-of-service attacks, scanning, spoofing, and brute-force attempts. Identifying malicious traffic manually can be difficult, especially when network traffic contains many records and different statistical characteristics.

This project addresses the problem by developing and evaluating a machine-learning-based classifier that distinguishes normal IoT network traffic from malicious traffic. The system will use labeled network traffic records to learn patterns associated with normal and attack activity. The project is an educational prototype. It will evaluate traffic records from a public dataset and a simulated stream; it will not directly monitor or protect a real operational network.

3. Research Question

How good can machine learning classify normal and malicious IoT network traffic using the Merged01 CSV from the CICIoT2023 dataset?

This will be determined by measuring accuracy, precision, recall, F1 score, and confusion matrix. However, recall and F1 score are more significant since they relate to missed malicious traffic.

4. Related Works

- **CICIoT2023:** CICIoT2023 is a dataset that Neto et al. (2023) created for large-scale attacks within an IoT environment. The dataset has been gathered through the use of a network topology consisting of 105 IoT devices and 33 attacks which were divided into seven categories, and CSV features have been extracted from the dataset for the purpose of conducting machine learning experiments.

- **UNSW-NB15:** Moustafa and Slay (2015) introduced UNSW-NB15, a network intrusion detection dataset containing normal activities and nine categories of attack traffic. It provides labeled network features and has been used in intrusion detection research. This work demonstrates the importance of labeled network datasets for developing and evaluating machine-learning-based security systems.

- **BoT-IoT:** Koroniotis et al. (2019) introduced the BoT-IoT dataset, which was designed under a real-world IoT network setting that had both benign and botnet network traffic. This study is pertinent since it looks at specific cyberattacks in IoT networks, like DoS, DDoS, scanning, keylogging, and data exfiltration attacks.

5. Machine Learning Plan

The project will use supervised binary classification.

Target label: Network traffic category.

Classes: Normal = 0 , Attack = 1.

Planned input features: Numeric network-traffic characteristics available in the selected CSV file. Candidate feature groups include packet and flow statistics, packet-size measurements, timing or inter-arrival statistics, traffic rates, and protocol or flag indicators.

Feature selection: The selected CSV file will be inspected first. The team will confirm the actual column names, data types, and label field before choosing the final input features. Label columns and identifiers that could reveal the answer will not be used as model inputs.

The initial model will be Logistic Regression as a baseline. Random Forest and Gradient Boosting will be evaluated in a later phase, followed by an SVM model in the final phase. The models will be compared using appropriate classification metrics.

6. Dataset and Data Plan

The project will use the CICIoT2023 dataset, specifically the merged CSV file containing normal and attack network traffic. The selected file, Merged01.csv, contains approximately 712,311 records and 40 columns, including 39 numerical network traffic features and one target label column (`Label`). Its file size is approximately 147 MB.

The dataset contains normal traffic (`BENIGN`) and 33 different attack categories, including DoS, DDoS, Mirai, reconnaissance, spoofing, and other attacks. For this project, the original labels will be converted into two classes: Normal (0) and Attack (1).

The team plans to use an initial balanced sample of approximately 10,000 records, consisting of 5,000 normal records and 5,000 attack records selected from different attack categories. Data preprocessing will include handling missing values, infinite values, and duplicate records before model training and evaluation.

7. Tools and Technologies

- Python 3: Programming language.
- Google Colab: Development of machine learning codes.
- GitHub: Repository of the project.
- Pandas and NumPy: Loading and preprocessing of the data.
- Scikit-learn: Machine Learning models and analysis.
- Matplotlib and Seaborn: Data visualizations.
- Streamlit: Dashboard development for final presentation.

8. Expected Outcome

The expected outcome is a working educational prototype that can classify selected IoT network traffic records as normal or malicious. The team will compare multiple machine-learning algorithms and evaluate their performance using classification metrics. A Python based stream simulator and a simple dashboard are planned for the final demonstration.

9. References

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